不定积分与定积分习题

1、 求下列一元不定积分

(1) $\int \frac{dx}{x^2}$

解 $\int \frac{dx}{x^2} = \int x^{-2}dx = -\frac{1}{x} + C$

(2) $\int \frac{dx}{\sqrt{x}}$

解 $\int \frac{dx}{\sqrt{x}} = \int x^{-\frac{1}{2}}dx = 2\sqrt{x} + C$

(3) $\int \sqrt[m]{x^n}dx$(m>0, n>0 为常数)

解 $\int \sqrt[m]{x^n}dx = \int x^{\frac{n}{m}}dx = \frac{m}{n+m}x^{\frac{n+m}{m}} + C$

(4) $\int (x^2 - 3x + 2)dx$

解 $\int (x^2 - 3x + 2)dx = \frac{1}{3}x^3 - \frac{3}{2}x^2 + 2x + C$

(5) $\int \frac{dh}{\sqrt{gh}}$(g 是大于 0 的常数)

解 $\int \frac{dh}{\sqrt{gh}} = \frac{2}{\sqrt{g}}\sqrt{h} + C$

(6) $\int (2e^x + \frac{3}{x})dx$

解 $\int (2e^x + \frac{3}{x})dx = 2e^x + 3\ln|x| + C$

(7) $\int x^3\sqrt{2-x^4}dx$

解 $\int x^3\sqrt{2-x^4}dx = -\frac{1}{4}\int(2-x^4)^{\frac{1}{2}}d(2-x^4) = -\frac{1}{6}(2-x^4)^{\frac{3}{2}} + C$

(8) $\int \frac{\cos\frac{3}{x}}{x^2}dx$

解 $\int \frac{\cos\frac{3}{x}}{x^2}dx = -\frac{1}{3}\int \cos\frac{3}{x}d\left(\frac{3}{x}\right) = -\frac{1}{3}\sin\frac{3}{x} + C$

(9) $\int \frac{e^{4\sqrt{x}}}{\sqrt{x}}dx$

解 $\int \frac{e^{4\sqrt{x}}}{\sqrt{x}}dx = \frac{1}{2}\int e^{4\sqrt{x}}d\left(4\sqrt{x}\right) = \frac{1}{2}e^{4\sqrt{x}} + C$

(10) $\int \frac{1}{x(1-2\ln x)}dx$

解 $\int \frac{1}{x(1-2\ln x)}dx = -\frac{1}{2}\int \frac{1}{1-2\ln x}d(1-2\ln x)$

$$= -\frac{1}{2}\ln|1-2\ln x| + C$$

(11) $\int e^x \sin(e^x+4)\,dx$

解 $\int e^x \sin(e^x+4)\,dx = \int \sin(e^x+4)\,d(e^x+4)$

$$= -\cos(e^x+4) + C$$

(12) $\int\left(\frac{3}{1+x^2}-\frac{2}{\sqrt{1-x^2}}\right)dx$

解 $\int\left(\frac{3}{1+x^2}-\frac{2}{\sqrt{1-x^2}}\right)dx = 3\arctan x - 2\arcsin x + C$

(13) $\int 3xe^x dx$

解 $\int 3xe^x dx = 3xe^x - \int 3e^x\,dx = 3xe^x - 3e^x + C$

(14) $\int \sec x(\sec x - \tan x)dx$

解 $\int \sec x(\sec x - \tan x)dx = \int(sec^2 x - \sec x \tan x)dx$

$= \tan x - \sec x + C$

(15) $\int cos^2\ \frac{x}{2}dx$

解 $\int cos^2\ \frac{x}{2}dx = \int \frac{\cos x+1}{2}dx = \frac{1}{2}\sin x + \frac{1}{2}x + C$

(16) $\int \frac{dx}{1+\cos 2x}$

解 $\int \frac{dx}{1+\cos 2x} = \int \frac{dx}{2cos^2 x} = \int \frac{1}{2}sec^2 x\,dx = \frac{1}{2}\tan x + C$

(17) $\int \frac{\cos 2x}{\cos x-\sin x}dx$

解 $\int \frac{\cos 2x}{\cos x-\sin x}dx = \int \frac{cos^2 x-sin^2 x}{\cos x-\sin x}dx = \int(\cos x + \sin x)dx$

$= \sin x - \cos x + C$

(18) $\int cot^2x\,dx$

解 $\int cot^2x\,dx = \int(cot^2x + 1 - 1)dx = \int(csc^2x - 1)dx$

$$= -cot\,x - x + \mathrm{C}$$

(19) $\int \cos\theta(\tan\theta + \sec\theta)d\theta$

解 $\int \cos\theta(\tan\theta + \sec\theta)d\theta = \int(\sin\theta + 1)d\theta = -\cos\theta + \theta + C$

(20) $\int \frac{3x^4+2x^2}{x^2+1}dx$

解 $\int \frac{3x^4+2x^2}{x^2+1}dx = \int \frac{3x^4+3x^2-x^2-1+1}{x^2+1}dx = \int\left(3x^2 - 1 + \frac{1}{x^2+1}\right)dx$

$$= x^3 - x + \arctan x + C$$

(21) $\int \frac{x}{\sqrt{2-3x^2}}dx$

解 $\int \frac{x}{\sqrt{2-3x^2}}dx = \int \frac{1}{2\sqrt{2-3x^2}}d(x^2) = -\frac{1}{3}\sqrt{2-3x^2} + C$

(22) $\int \frac{3x^3}{1-x^4}dx$

解 $\int \frac{3x^3}{1-x^4}dx = \int \frac{3}{4}\frac{1}{1-x^4}d(x^4) = -\frac{3}{4}\ln|1-x^4| + C$

(23) $\int \frac{\sin x}{cos^3x}dx$

解 $\int \frac{\sin x}{cos^3x}dx = \int -\frac{1}{cos^3x}d(\cos x) = \frac{1}{2cos^2x} + C$

(24) $\int \frac{\sin x+\cos x}{\sqrt[3]{\sin x-\cos x}}dx$

解 $\int \frac{\sin x+\cos x}{\sqrt[3]{\sin x-\cos x}}dx = \int \frac{1}{\sqrt[3]{\sin x-\cos x}}d(\sin x-\cos x)$

$$= \frac{3}{2}\sqrt[3]{(\sin x-\cos x)^2}+C$$

(25) $\int cos^2 x sin^5 x\, dx$

解 $\int cos^2 x sin^5 x\, dx = \int cos^2 x(1-cos^2 x)\sin x\, dx$

$$= -\int (cos^2 x - 2cos^4 x + cos^6 x)d(\cos x)$$

$$= -\frac{1}{3}cos^3 x + \frac{2}{5}cos^5 x - \frac{1}{7}cos^7 x + C$$

(26) $\int sec^6 x\, dx$

解 $\int sec^6 x\, dx = \int (1+2tan^2 x + tan^4 x)sec^2 x dx$

$$= \int (1+2tan^2 x + tan^4 x)d(\tan x)$$

$$= \tan x + \frac{2}{3}tan^3 x + \frac{1}{5}tan^5 x + C$$

(27) $\int \frac{1}{\arcsin x\cdot\sqrt{1-x^2}}dx$

解 $\int \frac{1}{\arcsin x\cdot\sqrt{1-x^2}}dx = \int \frac{1}{\arcsin x}d(\arcsin x) = \ln|\arcsin x| + C$

(28) $\int \tan\sqrt{1+x^2}\cdot\frac{xdx}{\sqrt{1+x^2}}$

解 $\int \tan\sqrt{1+x^2}\cdot\frac{xdx}{\sqrt{1+x^2}} = \int \tan\sqrt{1+x^2}\cdot\frac{d(x^2+1)}{2\sqrt{1+x^2}}$

$$= \int \tan\sqrt{1+x^2}\, d\left(\sqrt{1+x^2}\right) = -\ln\left|cos\sqrt{1+x^2}\right| + C$$

(29) $\int \frac{dx}{\sin x \cos x}$

解 $\int \frac{dx}{\sin x \cos x} = \int \frac{2}{\sin(2x)} dx = \int \frac{1}{\sin(2x)} d(2x)$

$$= \ln|\csc 2x - \cot 2x| + C = \ln|\tan x| + C$$

(30) $\int xsin\, xdx$

解 $\int xsin\, xdx = -xcos\, x + \int \cos x\, dx = -xcos\, x + \sin x + C$

(31) $\int \ln x\, dx$

解 $\int \ln x\, dx = xln\, x - \int dx = xln\, x - x + C$

(32) $\int xe^{-x}\, dx$

解 $\int xe^{-x}\, dx = -xe^{-x} + \int e^{-x}\, dx = -xe^{-x} - e^{-x} + C$

(33) $\int xtan^2 x\, dx$

解 $\int xtan^2 x\, dx = \int (xsec^2 x - x)\, dx$

$$= xtan\, x - \int \tan x dx - \int x\, dx = xtan\, x + \ln|\cos x| - \frac{1}{2}x^2 + C$$

(34) $\int \arcsin x\, dx$

解 $\int \arcsin x\, dx = xarcsin\, x - \int \frac{x}{\sqrt{1-x^2}} dx$

$$= xacrsin\, x + \sqrt{1-x^2} + C$$

2、 求下列一元定积分

(1) $\int_0^a (3x^2 - x + 1)\, dx$

解 $\int_0^a (3x^2 - x + 1)\, dx = [x^3 - \frac{1}{2}x^2 + x]_0^a = a^3 - \frac{1}{2}a^2 + a$

(2) $\int_1^2 (x^2 + \frac{1}{x^4})\, dx$

解 $\int_1^2 (x^2 + \frac{1}{x^4})\, dx = [\frac{1}{3}x^3 - \frac{1}{3}\frac{1}{x^3}]_1^2 = \frac{21}{8}$

(3) $\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{dx}{1+x^2}$

解 $\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{dx}{1+x^2} = [\arctan x]_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} = \frac{\pi}{6}$

(4) $\int_0^{\sqrt{3}a} \frac{dx}{a^2+x^2}$

解 $\int_0^{\sqrt{3}a} \frac{dx}{a^2+x^2} = [\frac{1}{a}(\arctan \frac{x}{a})]_0^{\sqrt{3}a} = \frac{\pi}{3a}$

(5) $\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{dx}{\sqrt{1-x^2}}$

解 $\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{dx}{\sqrt{1-x^2}} = [\arcsin x]_{-\frac{1}{2}}^{\frac{1}{2}} = \frac{\pi}{3}$

(6) $\int_0^1 \frac{dx}{\sqrt{4-x^2}}$

解 $\int_0^1 \frac{dx}{\sqrt{4-x^2}} = [\arcsin \frac{x}{2}]_0^1 = \frac{\pi}{6}$

(7) $\int_{-1}^0 \frac{3x^4+3x^2+1}{x^2+1} dx$

解 $\int_{-1}^0 \frac{3x^4+3x^2+1}{x^2+1} dx = \int_{-1}^0 \left(3x^2 + \frac{1}{1+x^2}\right) dx = [x^3 + \arctan x]_{-1}^0$

$$= \frac{\pi}{4} + 1$$

(8) $\int_0^{\frac{\pi}{4}} tan^2\theta \, d\theta$

解 $\int_0^{\frac{\pi}{4}} tan^2\theta \, d\theta = \int_0^{\frac{\pi}{4}} (sec^2\theta - 1) d\theta = [\tan\theta - \theta]_0^{\frac{\pi}{4}} = 1 - \frac{\pi}{4}$

(9) $\int_0^{2\pi} |\sin x| \, dx$

解 $\int_0^{2\pi} |\sin x| \, dx = \int_0^{\pi} \sin x \, dx + \int_{\pi}^{2\pi} -\sin x \, dx$

$$= [-\cos x]_0^{\pi} + [\cos x]_{\pi}^{2\pi} = 4$$

(10) $\int_0^2 f(x)dx$，其中$f(x) = \begin{cases} x+1, x \le 1, \\ \frac{1}{2}x^2, x > 1. \end{cases}$

解 $\int_0^2 f(x)dx = \int_0^1 (x+1)dx + \int_1^2 \frac{1}{2}x^2 dx$

$$= [\frac{1}{2}x^2 + x]_0^1 + [\frac{1}{6}x^3]_1^2 = \frac{8}{3}$$

(11) $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} cos^2u\,du$

解 $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} cos^2u\,du = \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{\cos 2u+1}{2}\,du = [\frac{1}{4}\sin 2u + \frac{u}{2}]_{\frac{\pi}{6}}^{\frac{\pi}{2}} = \frac{\pi}{6} - \frac{\sqrt{3}}{8}$

(12) $\int_0^{\sqrt{2}} \sqrt{2-x^2}\,dx$

解 令$x = \sqrt{2}\sin\theta$, 则定义域变为$[0, \frac{\pi}{2}]$,

原式$= \int_0^{\frac{\pi}{2}} \sqrt{2}\cos\theta \cdot \sqrt{2}\cos\theta\,d\theta = \int_0^{\frac{\pi}{2}} 2cos^2\theta d\theta = \frac{\pi}{2}$

(13) $\int_0^1 te^{-\frac{t^2}{2}}\,dt$

解 $\int_0^1 te^{-\frac{t^2}{2}}\,dt = \frac{1}{2}\int_0^1 e^{-\frac{t^2}{2}}\,d(t^2) = \frac{1}{2}[-2e^{-\frac{t^2}{2}}]_0^1 = 1 - e^{-\frac{1}{2}}$

(14) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x\cos 2x\,dx$

解 $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x\cos 2x\,dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1-2sin^2x)\,d(\sin x)$

$$= [\sin x - \frac{2}{3}sin^3x]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{2}{3}$$

(15) $\int_0^{\pi} \sqrt{1+\cos 2x}\,dx$

解 $\int_0^{\pi} \sqrt{1+\cos 2x}\,dx = \int_0^{\pi} \sqrt{2cos^2x}\,dx$

$$= \sqrt{2}(\int_0^{\frac{\pi}{2}} \cos x\,dx + \int_{\frac{\pi}{2}}^{\pi} -\cos x dx) = 2\sqrt{2}$$

(16) $\int_1^{+\infty} \frac{dx}{x^4}$

解 $\int_1^{+\infty} \frac{dx}{x^4} = [-\frac{1}{3}x^{-3}]_1^{+\infty} = 0 - \left(-\frac{1}{3}\right) = \frac{1}{3}$

(17) $\int_0^{+\infty} e^{-ax}\,dx\ (a > 0)$

解 $\int_0^{+\infty} e^{-ax}\,dx = [-\frac{1}{a}e^{-ax}]_0^{+\infty} = \frac{1}{a}$